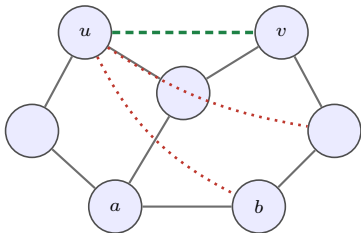


(a) Split and negative sampling



- observed edges (message passing)
- held-out positive edge (u, v)
- sampled negative pairs $(u, \cdot)$

(b) Ranking the positive against its negatives

predicted score $\hat{s}(u, \cdot)$		
rank 1	(u, n_3) neg	
rank 2	(u, v) pos	
rank 3	(u, b) neg	
⋮		

Hits@1 = 0, Hits@3 = 1

reciprocal rank = $\frac{1}{2}$

MRR averages this over all queries